

Introduction to Aggregating and Pivoting

DATA MANIPULATION IN KNIME



Philipp Kowalski
Certified KNIME Trainer

Introduction to Aggregation and Pivoting



Introduction to Aggregation and Pivoting



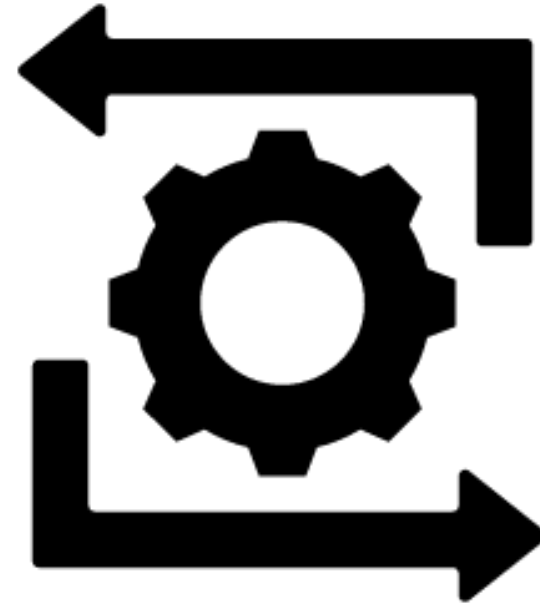
Introduction to Aggregation and Pivoting



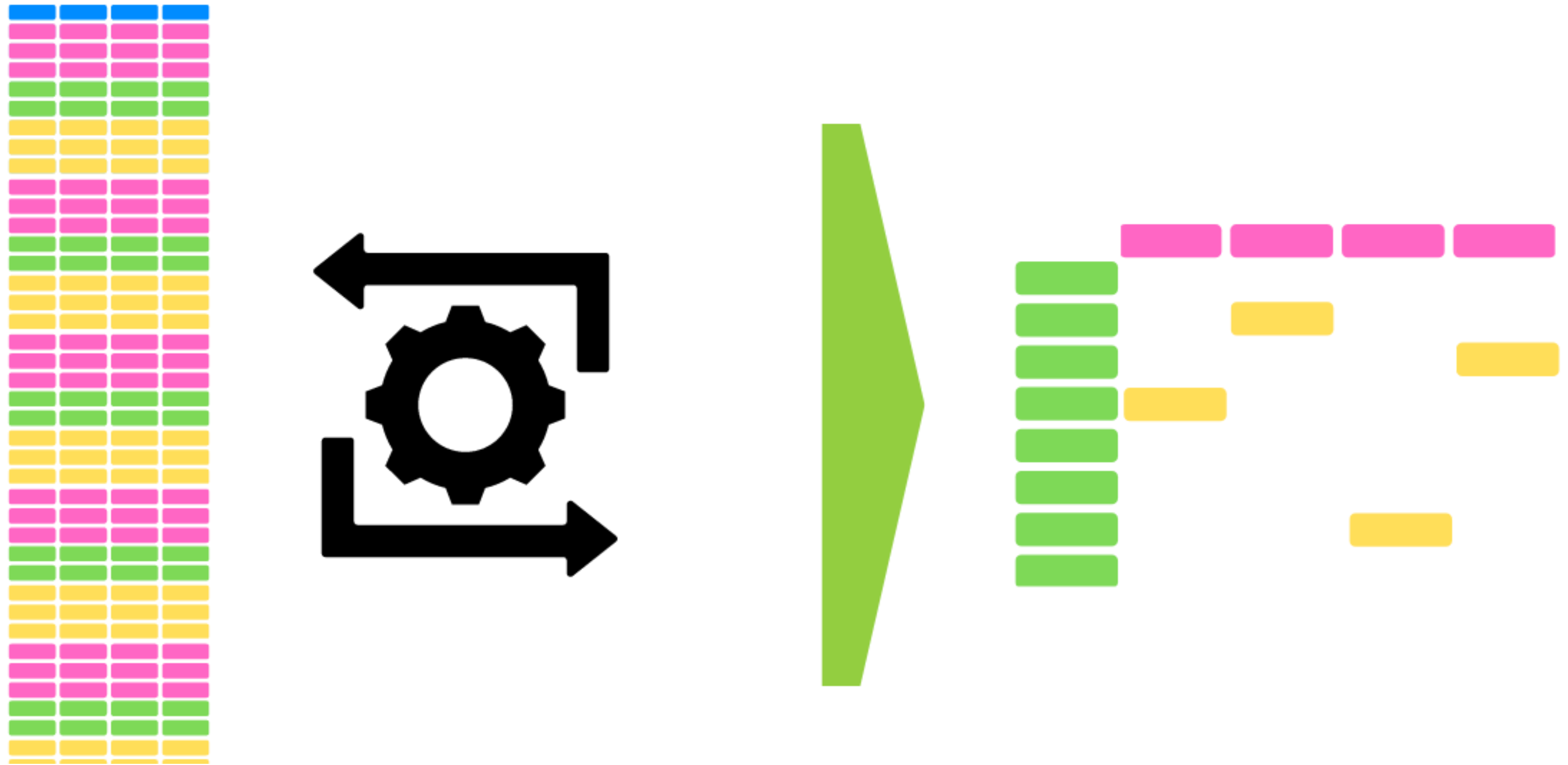
Introduction to Aggregation and Pivoting



Introduction to Aggregation and Pivoting

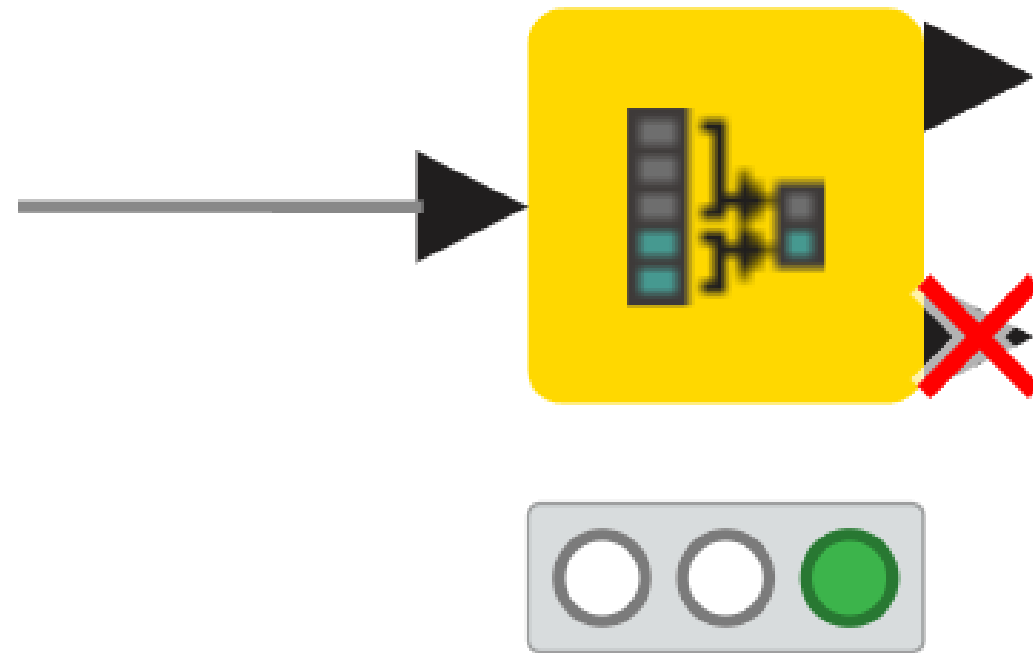


Introductions to Aggregation and Pivoting



Simple aggregations with the Row Aggregator

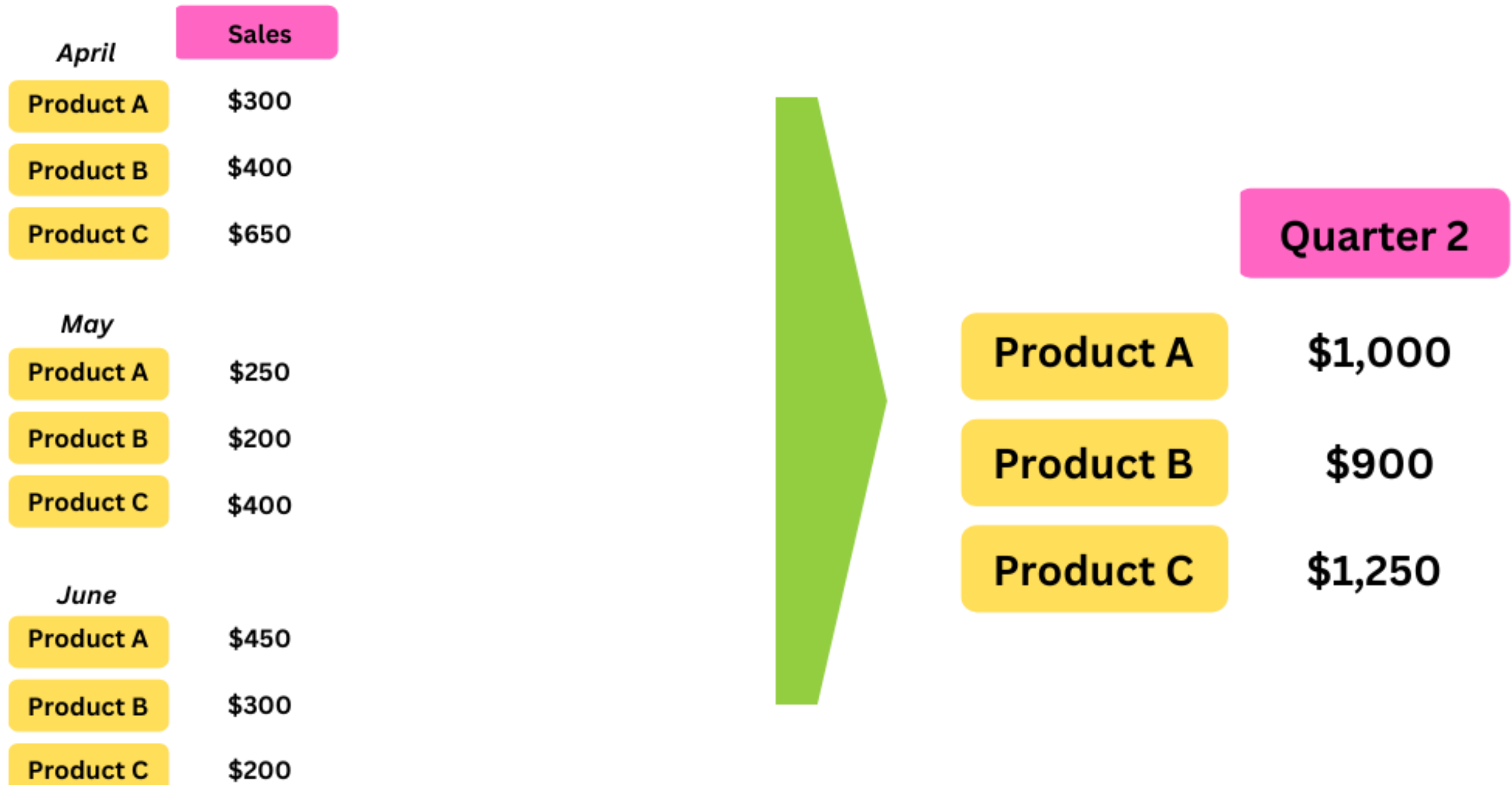
Row Aggregator



Simple aggregations with the Row Aggregator

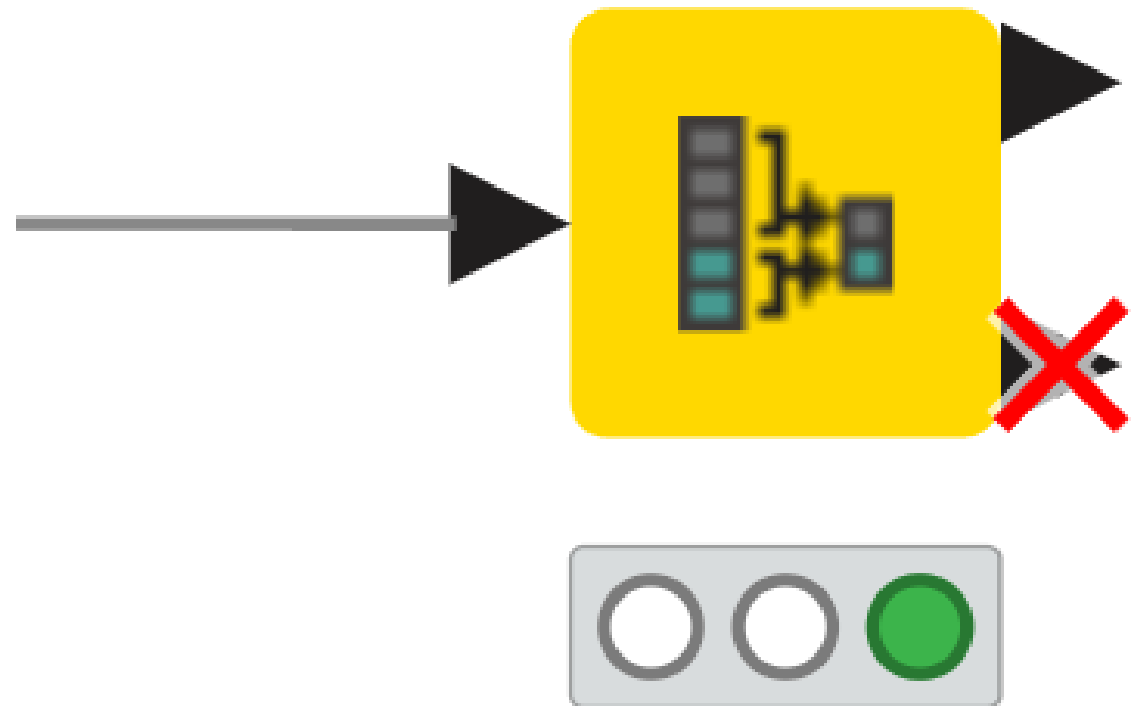
<i>April</i>	Sales
Product A	\$300
Product B	\$400
Product C	\$650
<i>May</i>	
Product A	\$250
Product B	\$200
Product C	\$400
<i>June</i>	
Product A	\$450
Product B	\$300
Product C	\$200

Simple aggregations with the Row Aggregator



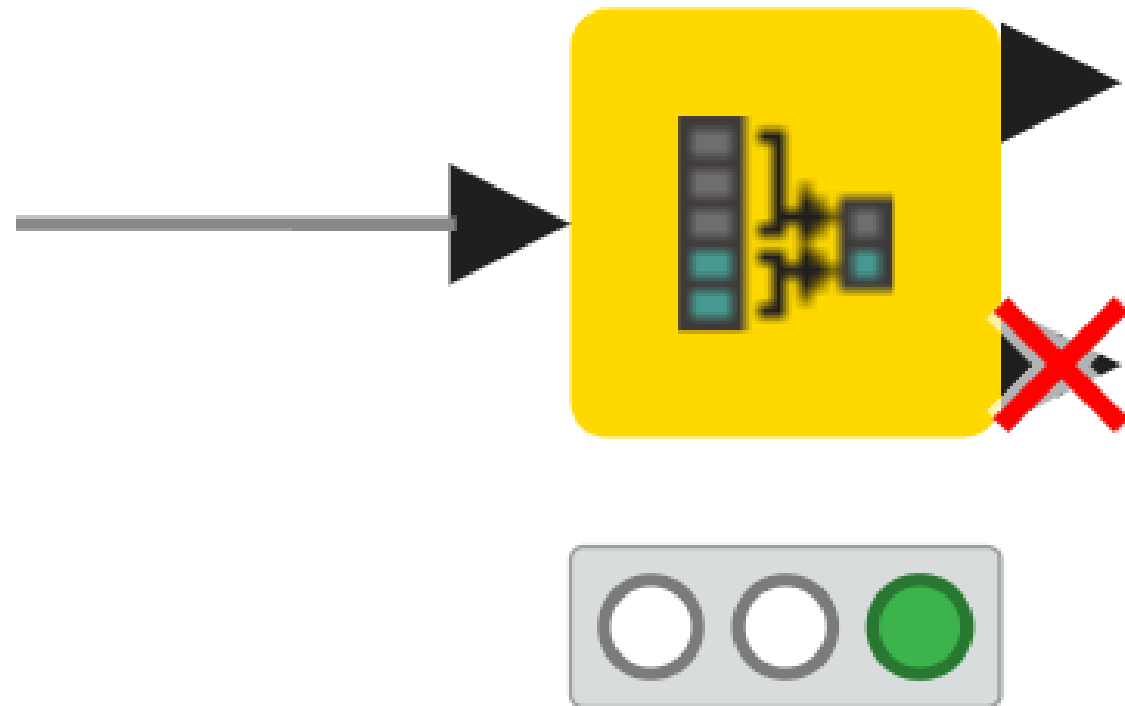
Simple aggregations with the Row Aggregator

Row Aggregator



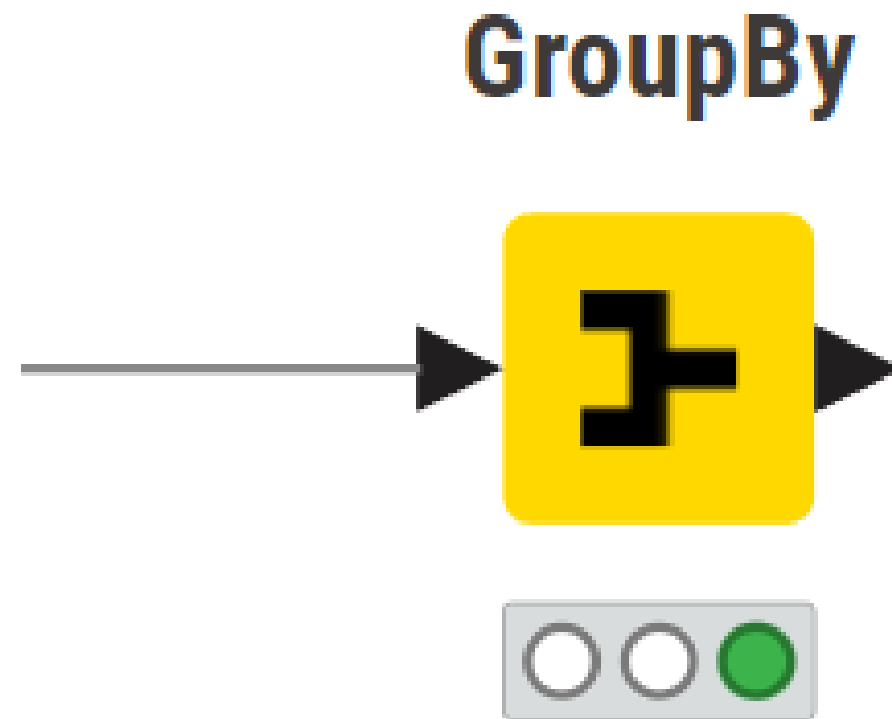
Simple aggregations with the Row Aggregator

Row Aggregator



- (Occurrence) Count
- Sum
- Average
- Minimum
- Maximum

Multi-level aggregations with GroupBy

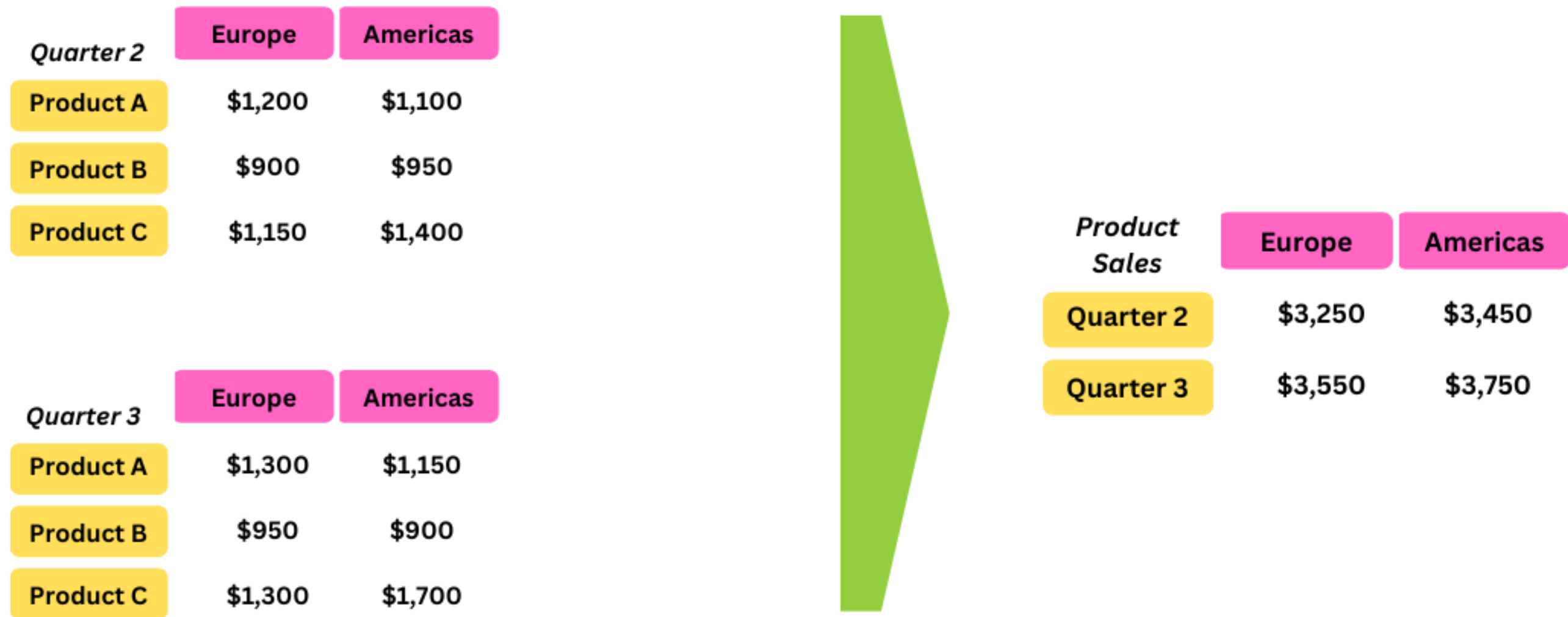


Multi-level aggregations with GroupBy

<i>Quarter 2</i>	Europe	Americas
Product A	\$1,200	\$1,100
Product B	\$900	\$950
Product C	\$1,150	\$1,400

<i>Quarter 3</i>	Europe	Americas
Product A	\$1,300	\$1,150
Product B	\$950	\$900
Product C	\$1,300	\$1,700

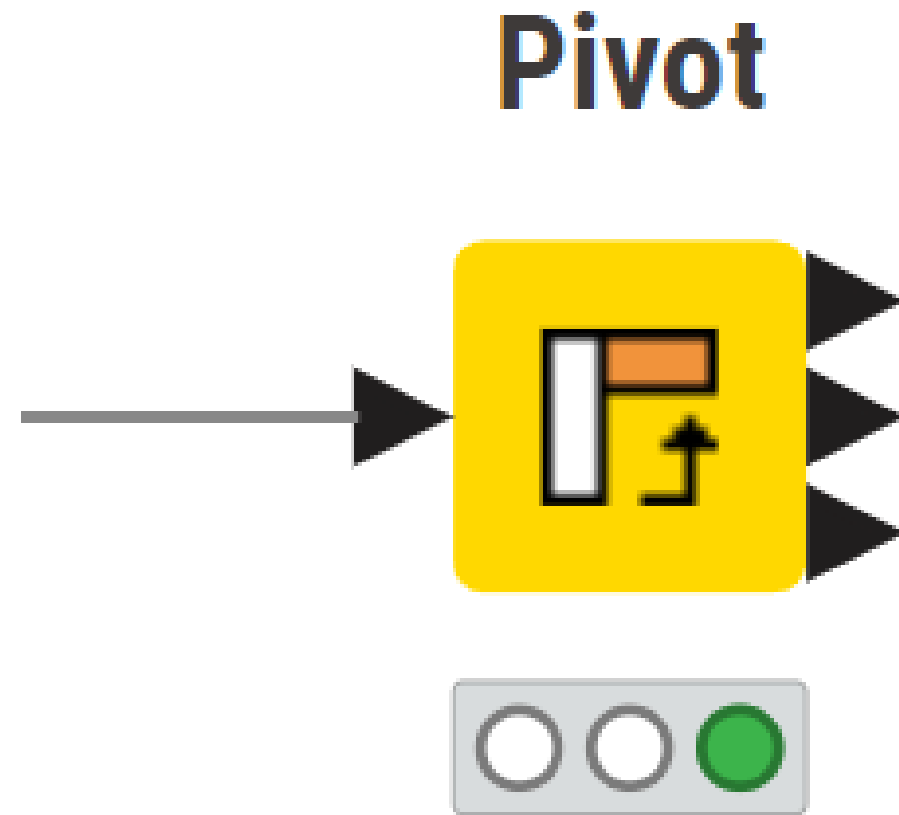
Multi-level aggregations with GroupBy



Multi-level aggregations with GroupBy



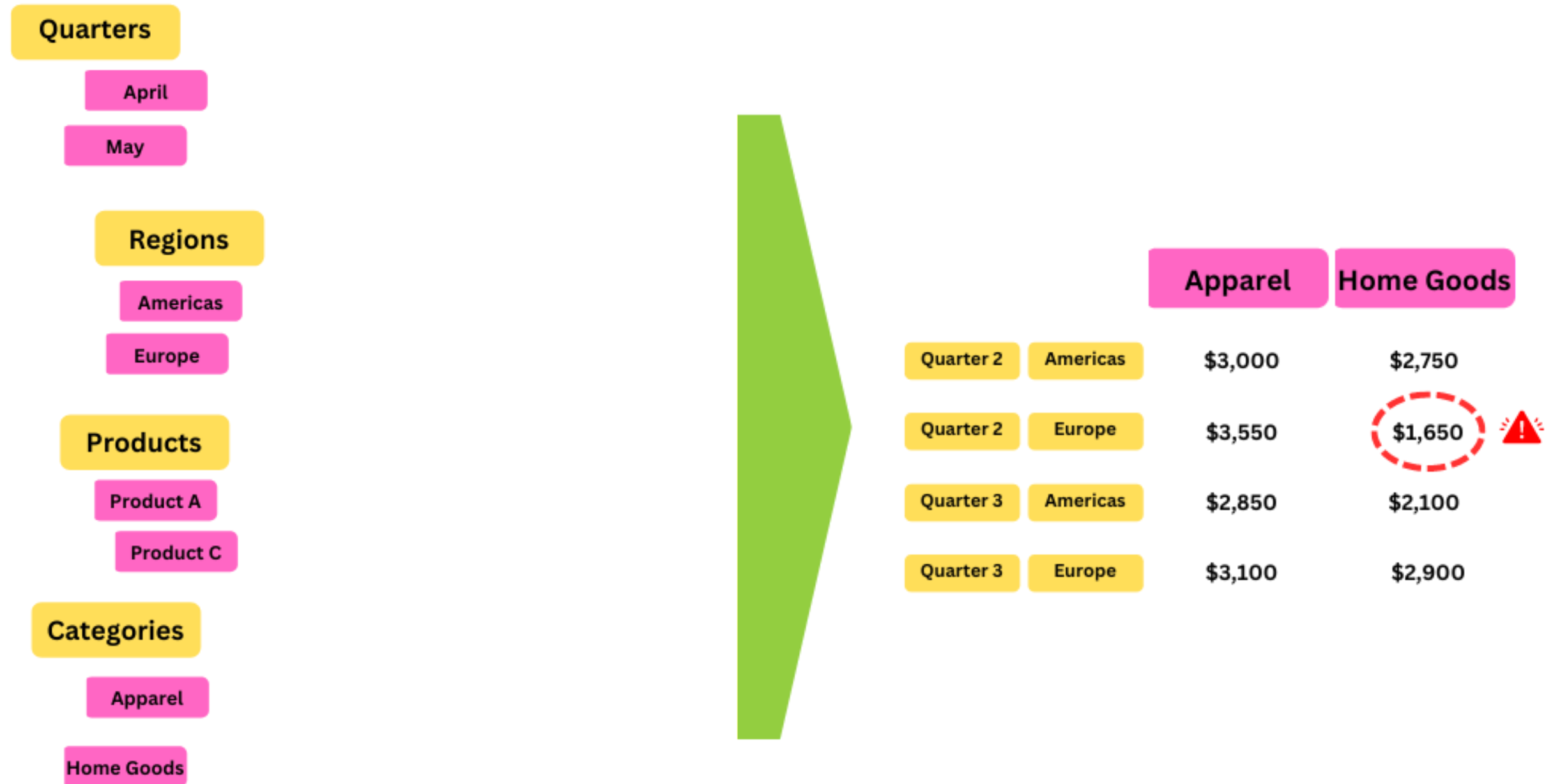
Complex aggregations with the Pivot node



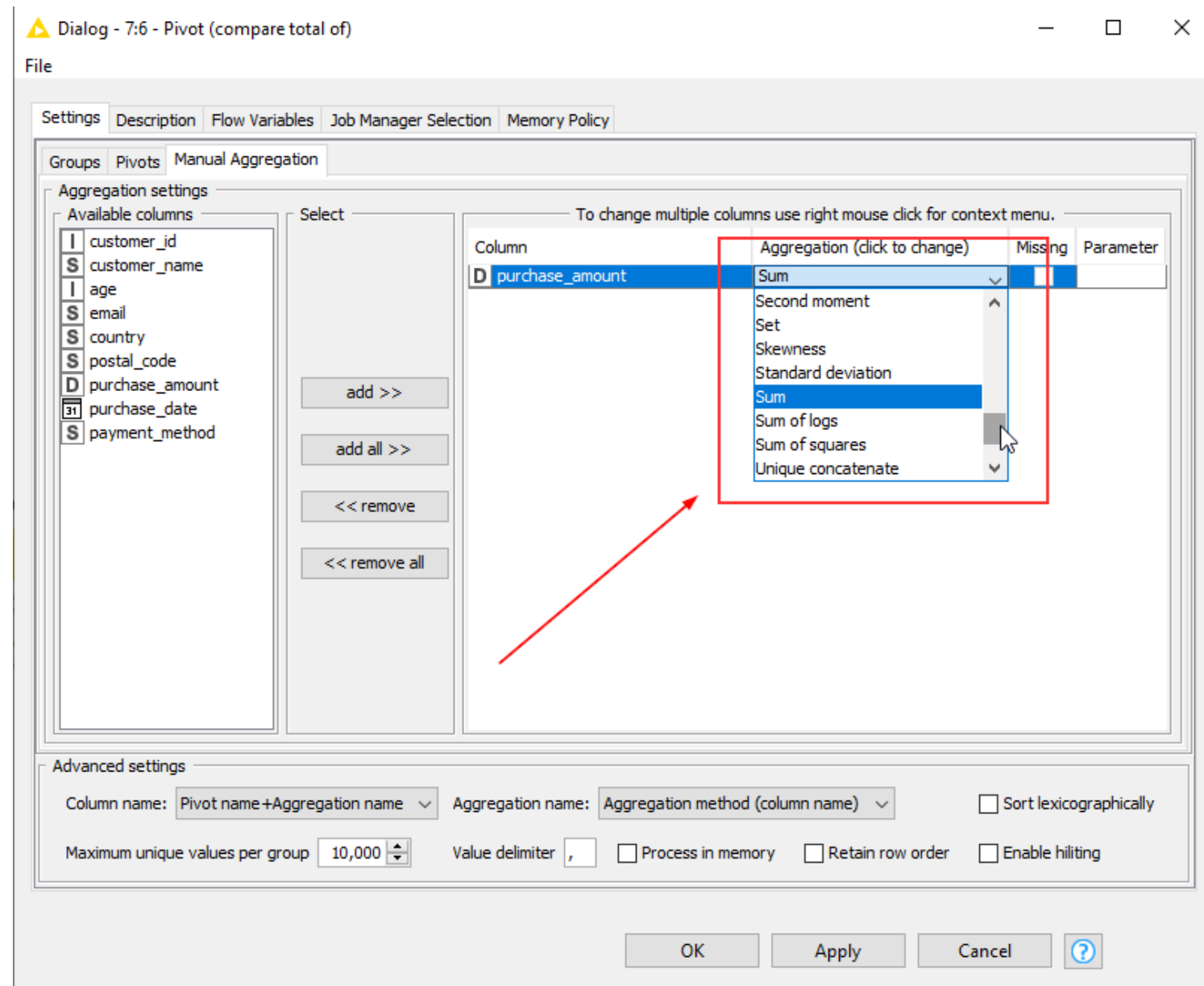
Complex aggregations with the Pivot node



Complex aggregations with the Pivot node



Complex aggregations with the Pivot node



Complex aggregations with the Pivot node

Dialog - 7:6 - Pivot (compare total of)

File

Settings Description Flow Variables Job Manager Selection Memory Policy

Groups Pivots Manual Aggregation

Aggregation settings

Available columns

- customer_id
- customer_name
- age
- email
- country
- postal_code
- purchase_amount
- purchase_date
- payment_method

Select

add >>

add all >>

<< remove

<< remove all

To change multiple columns use right mouse click for context menu.

Column	Aggregation (click to change)	Missing	Parameter
D purchase_amount	Sum	☐	

Second moment

Set

Skewness

Standard deviation

Sum

Sum of logs

Sum of squares

Unique concatenate

Advanced settings

Column name: Pivot name+Aggregation name

Aggregation name: Aggregation method (column name)

☐ Sort lexicographically

Maximum unique values per group: 10,000

Value delimiter: ,

☐ Process in memory

☐ Retain row order

☐ Enable hilling

OK Apply Cancel ?

Pivot table (Table)

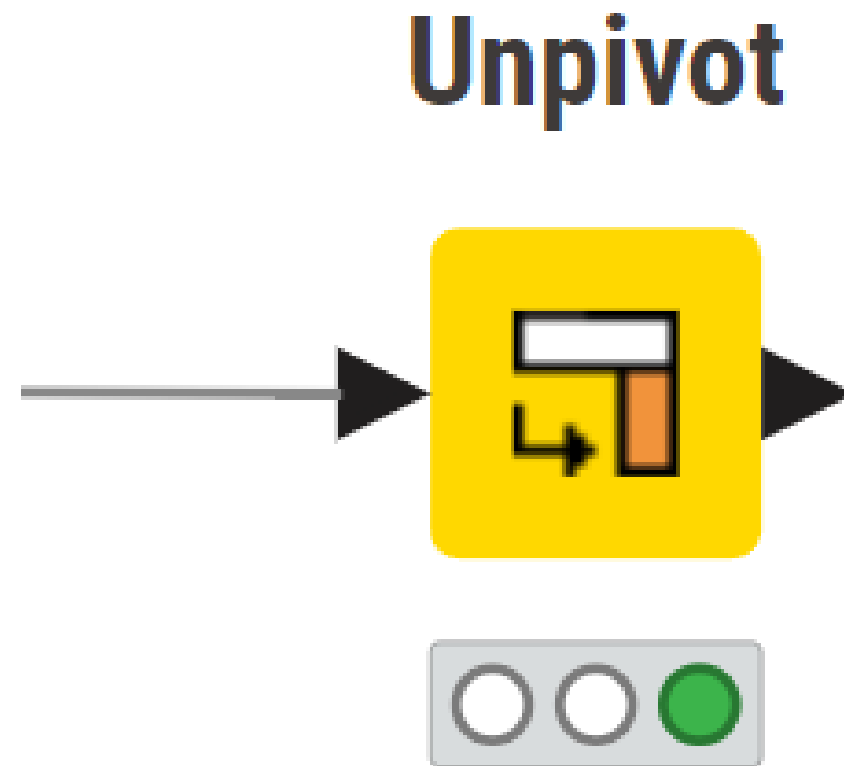
Rows: 12

|

Columns: 5

☐	#	RowID	Month (number) Number (integer)	Month (Name) String	clothing+Sum(purch... Number (double)	electronics+Sum(pur... Number (double)	home goods+Sum(p... Number (double)
☐	1	Row0	1	January	15,056.61	11,530.9	12,694.35
☐	2	Row1	2	February	14,152.12	13,255.38	9,814.33
☐	3	Row2	3	March	13,921.61	12,824.92	18,216.98
☐	4	Row3	4	April	16,525.78	21,793.2	15,475.6
☐	5	Row4	5	May	11,157.18	8,532.4	17,259.38
☐	6	Row5	6	June	18,965.8	10,650.66	18,665.12
☐	7	Row6	7	July	13,720.78	16,602.19	9,844.59
☐	8	Row7	8	August	12,935.55	14,563.27	11,741.15
☐	9	Row8	9	September	14,185.79	13,779.06	14,199.38
☐	10	Row9	10	October	15,408.02	12,660.83	14,614.42
☐	11	Row...	11	November	10,713.58	14,392.62	14,636.52
☐	12	Row...	12	December	13,187.37	15,329.57	14,057.04

Unpivoting data



Unpivoting data

		Apparel	Home Goods
Quarter 2	Americas	\$3,000	\$2,750
Quarter 2	Europe	\$3,550	\$1,650
Quarter 3	Americas	\$2,850	\$2,100
Quarter 3	Europe	\$3,100	\$2,900

Unpivoting data

		Apparel	Home Goods
Quarter 2	Americas	\$3,000	\$2,750
Quarter 2	Europe	\$3,550	\$1,650
Quarter 3	Americas	\$2,850	\$2,100
Quarter 3	Europe	\$3,100	\$2,900

Quarter	Region	Category	Sales
Quarter 2	Americas	Apparel	\$3,000
Quarter 3	Americas	Apparel	\$2,850
Quarter 2	Europe	Apparel	\$3,550
Quarter 3	Europe	Apparel	\$3,100
Quarter 2	Americas	Home Goods	\$2,750
Quarter 3	Americas	Home Goods	\$2,100
Quarter 2	Europe	Home Goods	\$1,650
Quarter 3	Europe	Home Goods	\$2,900

Let's practice!
DATA MANIPULATION IN KNIME

Aggregate data with GroupBy and Pivot nodes

DATA MANIPULATION IN KNIME



Philipp Kowalski
Instructor

Let's practice!
DATA MANIPULATION IN KNIME